DATA STRUCTURE IN C++ - T360

PROJECT-BASED LEARNING | PHASE-II

LibraMind

An Intelligent Digital Library and Personalized Book Recommendation System

PROJECT PROGRESS REPORT
Phase-II

Team: Code Crafters

Course: B.Tech - Computer Science Engineering

Academic year: 2026-2027

Mentor: Dr. Hemant Pokhariya

Graphic Era Deemed University
Dehradun, Uttarakhand

Project and Team Information

TEAM MEMBER	STUDENT ID	PROJECT RESPONSIBILITY
Ishan Pal (Team Lead)	2510012038	Frontend and student-side interface
Samiksha Saini	25101281166	Backend, database and admin functions
Vedansh Gusain	2510012637	DSA and recommendation system

Project Abstract

LibraMind is a digital library prototype that brings together core library operations and personalized book discovery. Students can use the interface to search and filter books, view details and availability, and access issue/return and account features. Librarians can manage book and student records through server-side functions backed by a database. A recommendation component uses book attributes and reading or borrowing activity to surface relevant titles. The work is divided across frontend, backend/database, and DSA/recommendation responsibilities. At this progress stage, the team reports the frontend, backend, and DSA work as completed; integration checks, evidence-based validation, and final reporting remain important close-out activities. The report distinguishes completed work from validation that still needs to be recorded.

Updated Project Approach and Architecture

The application follows a browser-to-server architecture. HTML, CSS, and JavaScript provide the student-facing interface. A Python Flask backend receives application requests, applies library rules, and communicates with the database. The database stores student and book records, availability, and issue/return transactions. The recommendation component uses book metadata and user activity; DSA techniques described for the project include hashing for lookup, sorting for rating/popularity order, a queue for a waiting list, a stack for recently viewed books, and graph relationships among books or subjects. The planned request path is: Student/Admin interface -> Flask routes or APIs -> database and recommendation logic -> results returned to the interface. The Phase-I proposal named MySQL or SQLite and the Book-Crossing dataset; the exact database, libraries, protocol details, and dataset usage in the current build should be confirmed against the implementation before submission.

Tasks Completed

The following work is reported as completed by the team. The assignment records the primary owner for clarity; integration may involve all members.

COMPLETED TASK	OWNER	SCOPE / OUTPUT
Frontend and student module	Ishan Pal	Website structure and styling; student-side flows including search/filter, book details, availability, and issue/return or dashboard screens as implemented.
Backend and database module	Samiksha Saini	Flask server-side work and data-management functions for books, students, transactions, and admin operations as implemented.
DSA and recommendation module	Vedansh Gusain	Recommendation and DSA component covering relevant book attributes and the selected search, sorting, hashing, queue, stack, and graph concepts.
Phase-I proposal and planning	All members	Defined LibraMind problem, intended users, proposed architecture, feature goals, and initial milestones.

Team Contribution Statements

> ***Ishan:*** *"My main role is frontend development and student-side functionality. I created the user interface and worked on book search, details, availability, issue/return, and dashboard features."*

> ***Samiksha:*** *"My main role is backend and database. I worked on server-side logic, data records, transactions, and admin functions."*

> ***Vedansh:*** *"My main role is the DSA and recommendation part. I worked on search, sorting, hashing, graph concepts, and recommendations for relevant books."*

Challenges and Roadblocks

The main challenge has been limited and fragmented development time because the team also had midterm examinations and college activities. This reduced the time available for coordinated integration and validation across the three modules. The team has divided ownership so members can progress in parallel, and will use focused integration sessions to connect the interface, Flask services, database, and recommendation component. The remaining work should be prioritized around a short end-to-end checklist: account flow, book search/details/availability, issue and return updates, admin record changes, and recommendation output. Any defects should be logged with an owner and resolved before the final demonstration. The report originality target is a similarity score below 20%. This is a target, not a measured result: the final submitted text should be checked using the institution-approved similarity checker, and any quoted or reused material should be attributed. This report paraphrases the Phase-I proposal rather than reproducing it at length.

Tasks Pending

PENDING TASK	OWNER	COMPLETION EVIDENCE
Connect and verify frontend, Flask backend, database, and recommendation flow end to end.	All members; Samiksha coordinates backend integration	Demonstrate the complete user journey and record the final interface/API/data path.
Run functional and edge-case checks for search, availability, issue/return, admin updates, and recommendations.	All members; each owner validates their module	Fill the testing table with actual pass/fail results and concise notes.
Confirm the final database, libraries, repository URL/branch, and key commits for this build.	Samiksha + team lead	Replace the report confirmation note with verified implementation details.
Run institutional similarity check and revise any overly similar passages.	All members	Record the actual similarity percentage only after running the checker; keep it below 20%.
Prepare final demonstration and report evidence (screenshots or a short walkthrough).	All members	Capture evidence from the working application and include it in the final submission if required.

Project Outcome and Deliverables

The intended outcome is a usable digital library prototype that combines student-facing discovery and circulation features with librarian controls and a recommendation component. Expected deliverables include a web interface, Flask application logic, a database for library records and transactions, DSA-backed search/ordering or activity structures, a recommendation output, and a working demonstration. The Phase-I concept and goals are retained, while the current report records the team completed module work and the remaining integration and verification steps.

Progress Overview

The team reports the three primary modules as completed: Ishan frontend, Samiksha backend/database, and Vedansh DSA/recommendation work. The project is behind the ideal schedule for integrated testing and final evidence because development time was constrained by midterm examinations and college activities. No work is claimed as ahead of schedule. The next priority is to connect the completed modules, verify the core flows, document actual test results, and run the institutional similarity checker. The overall completion percentage is not stated because no measured effort or milestone tracking data was supplied.

Module Flow

Member 1
Frontend + student module

THEN

Member 2
Flask backend + database/admin

THEN

Member 3
DSA + recommendation engine

THEN

LibraMind
Integrated intelligent digital library

Codebase Information

Repository link, active branch, and important commit identifiers were not included in the material supplied for this report. Add the verified repository URL, branch name, and key commits before submission. No repository details have been invented.

Testing and Validation Status

The source material did not provide test results. The table below deliberately records these as not yet documented; replace each row with the team actual result after testing. Do not mark a test as passed unless it was run against the current integrated build.

TEST TYPE	STATUS	NOTES / EVIDENCE TO RECORD
Student registration/login	Not documented	Verify valid and invalid inputs; record observed result.
Search, filter, details, availability	Not documented	Check matching results and current availability data.
Issue/return and transaction updates	Not documented	Confirm availability and transaction records update consistently.
Admin book/student management	Not documented	Check add, update, delete, and access control as implemented.
Recommendation and DSA behavior	Not documented	Use sample inputs; verify ranking/lookup behavior and explain limitations.
End-to-end integration	Pending	Run after the modules are connected; attach demonstration evidence.

Deliverables Progress

DELIVERABLE	STATUS	CURRENT NOTE
Frontend and student-side interface	Completed (reported)	Owned by Ishan Pal; confirm all listed flows in integrated build.
Backend, database, and admin functions	Completed (reported)	Owned by Samiksha Saini; confirm selected database and integration.
DSA and recommendation component	Completed (reported)	Owned by Vedansh Gusain; document tested behavior and examples.
Integrated prototype and functional validation	Pending	Complete end-to-end checks and record results.
Final report, codebase references, and similarity check	In progress	Insert verified repo details and actual similarity result below 20%.

Source note: Project identity, goals, and proposed architecture are summarized from the Phase-I report. Phase-II headings and word limits follow the supplied progress-report template. Sample names and placeholder text printed inside that template were treated as instructions/examples, not as project facts. Current completion details, team ownership, and scheduling constraints are based on the team information provided for this report.